01 | ACU geometry & instrumentation

Revision B · CAD-primary coordinates, dimensions, provenance and model checks

The source CAD now governs this package. Global coordinates replace the provisional PDF reconstruction; original text and PDF values remain traceable comparison records.

Deliverable content	Coverage
Physical sensor points	26 saved start/end datums across 15 sensors: 7 wall thermocouples, 4 penetrating thermocouples and 4 fiber starts.
Piping and useful locations	8 loop segments, 20 named locations, 2 main exit pipes, 6 analytical straight-face segments and 4 exact torus bends.
Coordinates and dimensions	Global X/Y/Z; signed segment Δx/Δy/Δz; centerline lengths; sensor projections, offsets and loop stations.
Machine-readable package	Schema 2.0 JSON, one CSV per table, pandas loader and numerical validation script. XLSX includes the same tables with linked calculations.
Model issue retained	Heater axial join gap = 0.433138 mm. Full tangent span includes this gap; no solid geometry was repaired.

Source: Annotated_ACU(2).FCStd — CAD extraction / saved annotations. Named datums and analytical cylinder / torus faces

02 | Source precedence & coordinate conventions

CAD global XYZ · all lengths in mm unless noted

Priority	Document name	Use
1	Annotated_ACU(2).FCStd	Primary geometry and named datums
2	ExtractedMeasurements.txt	Legacy source offsets / semantic hints
3	ACU_Detailed_General_Arrangement.pdf	Secondary drawing comparison
3	ACU_Detailed_General_Arrangement 2.pdf	Secondary detailed drawing comparison

Coordinate frame: native CAD X points right, Y points up in the front elevation, Z is out of the loop plane. No rotation or zeroing has been applied to exported data. Very small Z residuals are retained in JSON/CSV; printed coordinates are rounded to 0.001 mm, not tolerance specifications.

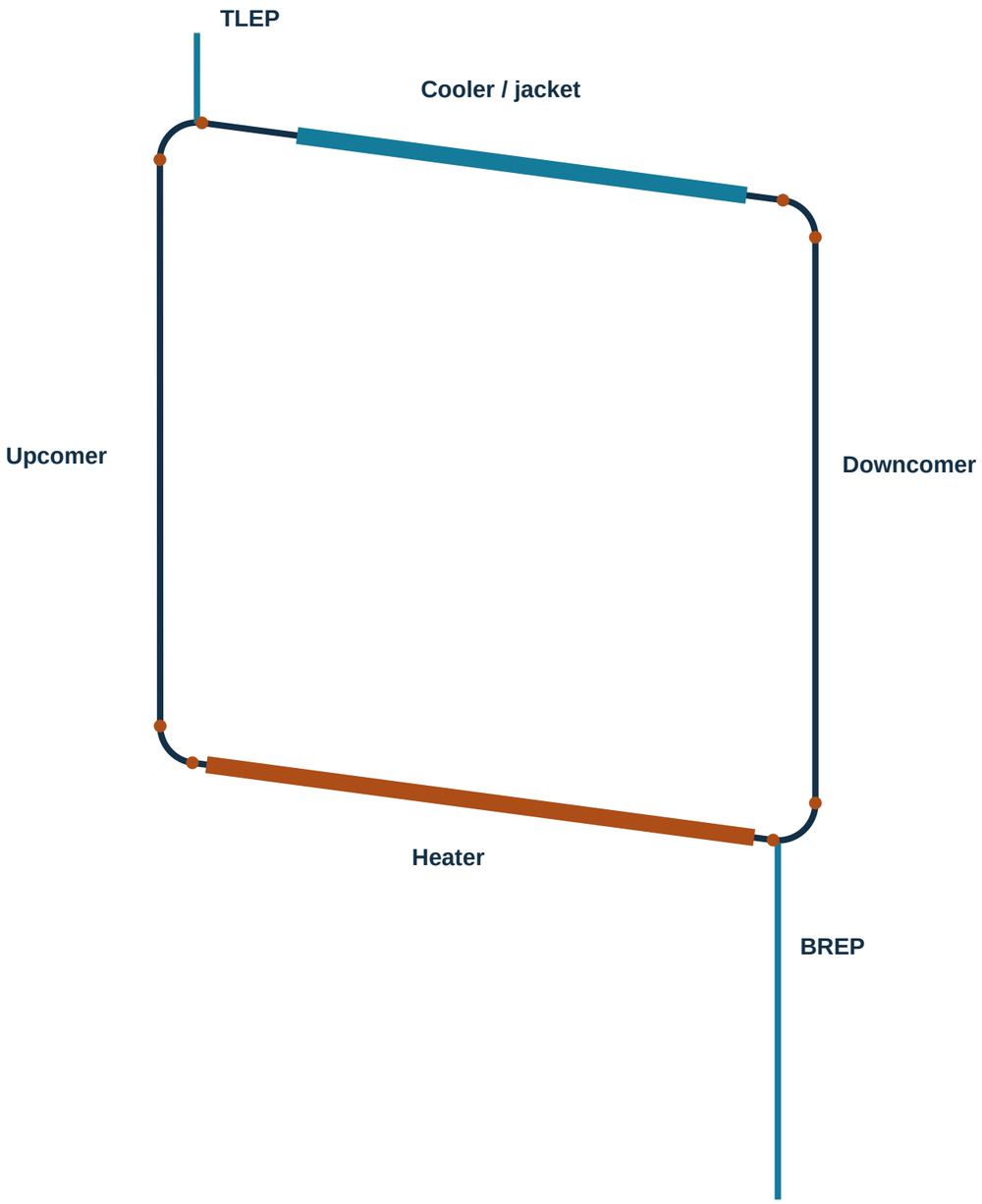
Local offsets: subtract HEATER_IN = (-10.614072, 856.270794, approximately 0) mm from a global point. Sensor axial offset u is instead measured along its own straight leg, from the upstream tangent. Station s starts at the DOWN_BOTTOM bend midpoint and increases Heater → Upcomer → Cooler → Downcomer.

Saved CAD Start_Point and End_Point labels identify annotation coordinates. They do not certify the active sensing junction, installation survey or DAQ channel. Fiber end points are undefined (null); a centerline projection is not a physical sensor coordinate.

Source: Annotated_ACU(2).FCStd — CAD extraction / saved annotations. Document.xml placements plus saved BRep geometry; no macro execution or recompute.

03 | Measured loop arrangement

Native CAD front elevation · uniform plot scale · tube centerlines, not fabrication outlines



The heater rises toward the upcomer at left. The cooler runs left-to-right and slopes downward into the downcomer. TLEP is upper left; BREP extends below the heater-side corner.

Colored overlays show heating tape and cooling-jacket axial extents. Exit lines terminate at their extracted cylindrical-face axis limits; they are not tank-port or junction-intersection coordinates.

Heater: L=1312.943; Δx=-1301.472; Δy=173.177 mm

Upcomer: L=1268.878; Δx=-0.628; Δy=1268.878 mm

Cooler: L=1313.723; Δx=1302.245; Δy=-173.280 mm

Downcomer: L=1267.684; Δx=0.000; Δy=-1267.684 mm

Source: Annotated_ACU(2).FCStd — CAD analytical faces and tangent datums. Full coordinate tables follow.

04 | Pipe start coordinates

CAD global XYZ · endpoints ordered in loop-flow direction · START tangent points

Segment	Component	Start location ID	X (mm)	Y (mm)	Z (mm)
BOTTOM	Heater	HEATER_IN	-10.614	856.271	0.000
BOTTOM_UP	Bottom-left bend	HEATER_OUT	-1312.086	1029.448	0.000
UP	Upcomer	UPCOMER_IN	-1384.287	1111.851	0.000
UP_TOP	Top-left bend	UPCOMER_OUT	-1384.915	2380.729	0.000
TOP	Cooler	COOLER_IN	-1290.773	2463.214	0.000
TOP_DOWN	Top-right bend	COOLER_OUT	11.472	2289.935	0.000
DOWN	Downcomer	DOWNCOMER_IN	84.150	2206.945	0.000
DOWN_BOTTOM	Bottom-right bend	DOWNCOMER_OUT	84.150	939.261	0.000

A bend begins at the previous straight tangent and ends at the next straight tangent. Straight spans include all collinear sections between named tangents; the heater gap is explicitly retained. See the Solid face boundaries sheet for individual material extents.

Source: Annotated_ACU(2).FCStd — CAD extraction / saved annotations. Named datums and analytical cylinder / torus faces

05 | Pipe end coordinates

CAD global XYZ · endpoints ordered in loop-flow direction · END tangent points

Segment	Component	End location ID	X (mm)	Y (mm)	Z (mm)
BOTTOM	Heater	HEATER_OUT	-1312.086	1029.448	0.000
BOTTOM_UP	Bottom-left bend	UPCOMER_IN	-1384.287	1111.851	0.000
UP	Upcomer	UPCOMER_OUT	-1384.915	2380.729	0.000
UP_TOP	Top-left bend	COOLER_IN	-1290.773	2463.214	0.000
TOP	Cooler	COOLER_OUT	11.472	2289.935	0.000
TOP_DOWN	Top-right bend	DOWNCOMER_IN	84.150	2206.945	0.000
DOWN	Downcomer	DOWNCOMER_OUT	84.150	939.261	0.000
DOWN_BOTTOM	Bottom-right bend	HEATER_IN	-10.614	856.271	0.000

A bend begins at the previous straight tangent and ends at the next straight tangent. Straight spans include all collinear sections between named tangents; the heater gap is explicitly retained. See the Solid face boundaries sheet for individual material extents.

Source: Annotated_ACU(2).FCStd — CAD extraction / saved annotations. Named datums and analytical cylinder / torus faces

06 | Segment differences & centerline lengths

Δ = end minus start · chord is endpoint distance · bend length is an arc, not its chord

Segment	Δx	Δy	Δz	Chord	CL length	Gap
BOTTOM	-1301.472	173.177	0.000	1312.943	1312.943	0.433
BOTTOM_UP	-72.201	82.403	0.000	109.559	119.601	0.000
UP	-0.628	1268.878	0.000	1268.878	1268.878	0.000
UP_TOP	94.141	82.485	0.000	125.165	141.688	0.000
TOP	1302.245	-173.280	0.000	1313.723	1313.723	0.000
TOP_DOWN	72.679	-82.990	0.000	110.316	120.434	0.000
DOWN	0.000	-1267.684	0.000	1267.684	1267.684	0.000
DOWN_BOTTOM	-94.764	-82.990	0.000	125.967	142.584	0.000

Segment	CAD extraction locator
BOTTOM	Part__Feature140.Face18 + Part__Feature143.Face13; tangent datums; axial gap retained
BOTTOM_UP	Part__Feature143.Face14
UP	Part__Feature143.Face15; tangent datums
UP_TOP	Part__Feature143.Face16
TOP	Part__Feature143.Face17 + Part__Feature140.Face14; tangent datums
TOP_DOWN	Part__Feature140.Face15
DOWN	Part__Feature140.Face16; tangent datums
DOWN_BOTTOM	Part__Feature140.Face17

Source: Annotated_ACU(2).FCStd — CAD extraction / saved annotations. Named datums and analytical cylinder / torus faces

07 | Named coordinate locations / 1 of 2

CAD global XYZ · all lengths in mm unless noted

Location ID	X	Y	Z	CAD datum / extraction reference
HEATER_IN	-10.614	856.271	0.000	Bottom_Leg_Start_Plane [DatumPlane]; support: Part__Feature140.Edge54
HEATER_OUT	-1312.086	1029.448	0.000	Bottom_Leg_End_Point [DatumPoint]; support: Part__Feature143.Edge39
UPCOMER_IN	-1384.287	1111.851	0.000	Up_Leg_Start_Plane [DatumPlane002]; support: Part__Feature143.Edge41
UPCOMER_OUT	-1384.915	2380.729	0.000	Up_Leg_End_Point [DatumPoint001]; support: Part__Feature143.Edge44
COOLER_IN	-1290.773	2463.214	0.000	Top_Leg_Start_Plane [DatumPlane003]; support: Part__Feature143.Edge46
COOLER_OUT	11.472	2289.935	0.000	Top_Leg_End_Point [DatumPoint002]; support: Part__Feature140.Edge46
DOWNCOMER_IN	84.150	2206.945	0.000	Down_Leg_Start_Plane [DatumPlane004]; support: Part__Feature140.Edge48
DOWNCOMER_OUT	84.150	939.261	0.000	Down_Leg_End_Point [DatumPoint003]; support: Part__Feature140.Edge52
JACKET_IN	-1077.480	2434.833	0.000	Cooling_Jacket_Start_Point [DatumPoint004]; support: Part__Feature182.Edge9
JACKET_OUT	-70.357	2300.823	0.000	Cooling_Jacket_End_Point [DatumPoint005]; support: Part__Feature181.Edge10

Source: Annotated_ACU(2).FCStd — CAD extraction / saved annotations. Named datums and analytical cylinder / torus faces

08 | Named coordinate locations / 2 of 2

CAD global XYZ · all lengths in mm unless noted

Location ID	X	Y	Z	CAD datum / extraction reference
TAPE_IN	-53.182	861.935	0.000	Heating_Tape_Start_Point [DatumPoint006]; support: Part__Feature146.Edge6
TAPE_OUT	-1280.613	1025.260	0.000	Heating_Tape_End_Point [DatumPoint007]; support: Part__Feature146.Edge7
HEATER_LONG_FACE_END	-1255.006	1021.852	0.000	Part__Feature140.Face18; cylinder axis_start
HEATER_SHORT_FACE_START	-1255.435	1021.910	0.000	Part__Feature143.Face13; cylinder axis_end
COOLER_SOLID_JOIN	-45.179	2297.473	0.000	Part__Feature143.Face17; cylinder axis_start
TLEP_LOOP_SIDE	-1301.784	2461.348	0.000	Part__Feature141.Face6; cylinder axis_start; axial face boundary, not flow intersection
TLEP_REMOTE_SIDE	-1301.884	2664.548	0.000	Part__Feature141.Face6; cylinder axis_end; axial face boundary, not flow intersection
BREP_LOOP_SIDE	0.000	850.900	0.000	Part__Feature142.Face96; cylinder axis_start; axial face boundary, not flow intersection
BREP_REMOTE_SIDE	0.000	50.800	0.000	Part__Feature142.Face96; cylinder axis_end; axial face boundary, not flow intersection
S0_BEND_MIDPOINT	55.587	876.277	0.000	Part__Feature140.Face17; half-sweep centerline point

Source: Annotated_ACU(2).FCStd — CAD extraction / saved annotations. Named datums and analytical cylinder / torus faces

09 | Analytical bend geometry

Exact torus major radius and trimmed parameter sweep govern arc length

Bend	Center X	Center Y	Radius R	Sweep (deg)	Arc R × sweep(rad)
BOTTOM_UP	-1301.116	1111.892	83.171	82.392	119.601
UP_TOP	-1301.744	2380.770	83.171	97.608	141.688
TOP_DOWN	0.429	2206.945	83.721	82.421	120.434
DOWN_BOTTOM	0.429	939.261	83.721	97.579	142.584

Bend	Source locator
BOTTOM_UP	Part__Feature143.Face14; torus major radius and trimmed U range
UP_TOP	Part__Feature143.Face16; torus major radius and trimmed U range
TOP_DOWN	Part__Feature140.Face15; torus major radius and trimmed U range
DOWN_BOTTOM	Part__Feature140.Face17; torus major radius and trimmed U range

Revision A retained the text code’s ambiguous angle convention. Revision B supersedes that calculation with measured CAD torus sweeps. The old values remain only in legacy_bends.csv, marked SUPERSEDED_REFERENCE_ONLY.

Source: Annotated_ACU(2).FCStd — CAD extraction / saved annotations. Named datums and analytical cylinder / torus faces

10 | Loop station definition & schedule

s=0 at DOWN_BOTTOM bend midpoint · increasing with the named loop sequence

Leg	Flat L	Entering ½ arc	Leaving ½ arc	Leg start s	Flat start s	Flat end s	Leg end s
BOTTOM	1312.943	71.292	59.801	0	71.292	1384.235	1444.036
UP	1268.878	59.801	70.844	1444.036	1503.836	2772.715	2843.559
TOP	1313.723	70.844	60.217	2843.559	2914.403	4228.126	4288.343
DOWN	1267.684	60.217	71.292	4288.343	4348.560	5616.244	5687.536

Total station length: 5687.535910 mm. The centerline sequence is closed, but this is not proof of a watertight solid: it spans the 0.433138 mm heater gap.

For a point Q on a sensor datum and a straight with endpoints A, B: d=(B−A)/|B−A|; u=(Q−A)·d; projected centerline point=A+u·d; sensor station s=flat_start_s+u. The radial offset is |Q−projection|.

Source: Annotated_ACU(2).FCStd — CAD extraction / saved annotations. Tangent endpoints, analytical torus half-arcs and named sensor datums.

11 | Wall thermocouple coordinates

Saved physical CAD datums · START and END are separate records · not centerline projections

Sensor / role	CAD label	X	Y	Z	CAD object
ACU_WT01 / start	WTC_1_Start_Point	-1189.674	1014.530	17.448	DatumPoint010
ACU_WT01 / end	WTC_1_End_Point	-1224.413	1019.152	17.448	DatumPoint011
ACU_WT02 / start	WTC_2_Start_Point	-1390.235	1401.991	16.510	DatumPoint015
ACU_WT02 / end	WTC_2_End_Point	-1390.252	1437.035	16.510	DatumPoint016
ACU_WT03 / start	WTC_3_Start_Point	-1390.627	2195.740	16.510	DatumPoint017
ACU_WT03 / end	WTC_3_End_Point	-1390.645	2230.785	16.510	DatumPoint018
ACU_WT04 / start	WTC_4_Start_Point	-1082.800	2442.033	17.680	DatumPoint022
ACU_WT04 / end	WTC_4_End_Point	-1048.062	2437.410	17.680	DatumPoint023
ACU_WT05 / start	WTC_5_Start_Point	-98.078	2311.003	17.680	DatumPoint024
ACU_WT05 / end	WTC_5_End_Point	-63.340	2306.381	17.680	DatumPoint025
ACU_WT06 / start	WTC_6_Start_Point	81.805	2050.993	17.342	DatumPoint032
ACU_WT06 / end	WTC_6_End_Point	81.805	2015.948	17.342	DatumPoint033
ACU_WT07 / start	WTC_7_Start_Point	81.799	1460.391	17.342	DatumPoint034
ACU_WT07 / end	WTC_7_End_Point	81.799	1425.346	17.342	DatumPoint035

Source: Annotated_ACU(2).FCStd — CAD extraction / saved annotations. Exact support edge references and verification statuses are included in sensor_points.csv / XLSX Sensor Points.

12 | Penetrating thermocouple coordinates

Saved physical CAD datums · START and END are separate records · not centerline projections

Sensor / role	CAD label	X	Y	Z	CAD object
ACU_PT01 / start	PTC_1_Start_Point	-26.078	869.772	-6.549	DatumPoint028
ACU_PT01 / end	PTC_1_End_Point	-27.065	862.358	-2.231	DatumPoint009
ACU_PT02 / start	PTC_2_Start_Point	-1298.001	1039.017	-6.549	DatumPoint029
ACU_PT02 / end	PTC_2_End_Point	-1299.044	1031.174	-1.981	DatumPoint014
ACU_PT03 / start	PTC_3_Start_Point	-1132.635	2453.616	-6.549	DatumPoint030
ACU_PT03 / end	PTC_3_End_Point	-1133.679	2445.773	-1.981	DatumPoint021
ACU_PT04 / start	PTC_4_Start_Point	-12.210	2304.530	-6.549	DatumPoint026
ACU_PT04 / end	PTC_4_End_Point	-13.979	2291.235	1.194	DatumPoint027

PTC_2 and PTC_3 End_Point saved coordinates match circle centers on Edge29 in their components, not the serialized support Edge38 in extracted BRep order. Coordinates are preserved; support-index interpretation remains flagged for review.

Source: Annotated_ACU(2).FCStd — CAD extraction / saved annotations. Exact support edge references and verification statuses are included in sensor_points.csv / XLSX Sensor Points.

13 | Fiber start coordinates

Saved physical CAD datums · START and END are separate records · not centerline projections

Sensor / role	CAD label	X	Y	Z	CAD object
ACU_FIB01 / start	LF_1_Start_Point	-1297.874	1042.474	0.000	DatumPoint012
ACU_FIB02 / start	LF_2_Start_Point	-1399.639	2253.721	0.000	DatumPoint019
ACU_FIB03 / start	LF_3_Start_Point	-12.220	2307.825	0.000	DatumPoint031
ACU_FIB04 / start	LF_4_Start_Point	98.937	1066.261	0.000	DatumPoint036

Only the four named fiber Start_Point datums are supplied. No endpoint, distributed trajectory or active fiber sensing-channel positions are inferred.

Source: Annotated_ACU(2).FCStd — CAD extraction / saved annotations. Exact support edge references and verification statuses are included in sensor_points.csv / XLSX Sensor Points.

14 | Sensor centerline projections & loop stations

These are derived axial locations, distinct from physical XYZ on the preceding sheets

Sensor	Leg	u from tangent	s around loop	CL X	CL Y	Radial offset
ACU_FIB01	BOTTOM	1300.573	1371.866	-1299.824	1027.816	14.787
ACU_FIB02	UP	1141.878	2645.715	-1384.852	2253.729	14.787
ACU_FIB03	TOP	1287.878	4202.281	-14.147	2293.343	14.609
ACU_FIB04	DOWN	1140.684	5489.244	84.150	1066.261	14.787
ACU_WT01	BOTTOM	1189.633	1260.926	-1189.854	1013.183	17.501
ACU_WT02	UP	290.143	1793.979	-1384.430	1401.993	17.501
ACU_WT03	UP	1083.893	2587.729	-1384.823	2195.743	17.501
ACU_WT04	TOP	208.950	3123.353	-1083.649	2435.654	18.815
ACU_WT05	TOP	1202.351	4116.754	-98.927	2304.624	18.815
ACU_WT06	DOWN	155.952	4504.512	84.150	2050.993	17.500
ACU_WT07	DOWN	746.553	5095.113	84.150	1460.391	17.501
ACU_PT01	BOTTOM	17.110	88.402	-27.575	858.528	13.099
ACU_PT02	BOTTOM	1300.243	1371.535	-1299.497	1027.772	13.099
ACU_PT03	TOP	158.023	3072.426	-1134.131	2442.371	13.099
ACU_PT04	TOP	1288.323	4202.726	-13.706	2293.285	13.099

Source: Annotated_ACU(2).FCStd — CAD extraction / saved annotations. Projection of Start_Point onto the named straight tangent axis. Z coordinates are retained in the data package.

15 | Heater sensor location detail

Numbered strip markers avoid overlapping labels; physical XYZ and projected axial spacing are tabulated



Tag	Sensor	u	Δu previous	X physical	Y physical	Z physical
1	ACU_PT01	17.110	—	-26.078	869.772	-6.549
2	ACU_WT01	1189.633	1172.523	-1189.674	1014.530	17.448
3	ACU_PT02	1300.243	110.610	-1298.001	1039.017	-6.549
4	ACU_FIB01	1300.573	0.330	-1297.874	1042.474	0.000

Sensor pair (start points)	Signed Δx	Signed Δy	Signed Δz	3D distance
ACU_PT01 → ACU_WT01	-1163.596	144.758	23.997	1172.811
ACU_WT01 → ACU_PT02	-108.326	24.487	-23.997	113.623
ACU_PT02 → ACU_FIB01	0.127	3.457	6.549	7.407

Source: Annotated_ACU(2).FCStd — CAD extraction / saved annotations. CAD Start_Point positions; ordering and Δu use centerline projection.

16 | Upcomer sensor location detail

Numbered strip markers avoid overlapping labels; physical XYZ and projected axial spacing are tabulated



Tag	Sensor	u	Δu previous	X physical	Y physical	Z physical
1	ACU_WT02	290.143	—	-1390.235	1401.991	16.510
2	ACU_WT03	1083.893	793.750	-1390.627	2195.740	16.510
3	ACU_FIB02	1141.878	57.985	-1399.639	2253.721	0.000

Sensor pair (start points)	Signed Δx	Signed Δy	Signed Δz	3D distance
ACU_WT02 → ACU_WT03	-0.393	793.750	0.000	793.750
ACU_WT03 → ACU_FIB02	-9.011	57.981	-16.510	60.956

Source: Annotated_ACU(2).FCStd — CAD extraction / saved annotations. CAD Start_Point positions; ordering and Δu use centerline projection.

17 | Cooler sensor location detail

Numbered strip markers avoid overlapping labels; physical XYZ and projected axial spacing are tabulated



Tag	Sensor	u	Δu previous	X physical	Y physical	Z physical
1	ACU_PT03	158.023	—	-1132.635	2453.616	-6.549
2	ACU_WT04	208.950	50.927	-1082.800	2442.033	17.680
3	ACU_WT05	1202.351	993.401	-98.078	2311.003	17.680
4	ACU_FIB03	1287.878	85.527	-12.220	2307.825	0.000
5	ACU_PT04	1288.323	0.445	-12.210	2304.530	-6.549

Sensor pair (start points)	Signed Δx	Signed Δy	Signed Δz	3D distance
ACU_PT03 → ACU_WT04	49.835	-11.583	24.230	56.610
ACU_WT04 → ACU_WT05	984.722	-131.029	0.000	993.401
ACU_WT05 → ACU_FIB03	85.858	-3.178	-17.680	87.717
ACU_FIB03 → ACU_PT04	0.010	-3.295	-6.549	7.332

Source: Annotated_ACU(2).FCStd — CAD extraction / saved annotations. CAD Start_Point positions; ordering and Δu use centerline projection.

18 | Downcomer sensor location detail

Numbered strip markers avoid overlapping labels; physical XYZ and projected axial spacing are tabulated



Tag	Sensor	u	Δu previous	X physical	Y physical	Z physical
1	ACU_WT06	155.952	—	81.805	2050.993	17.342
2	ACU_WT07	746.553	590.602	81.799	1460.391	17.342
3	ACU_FIB04	1140.684	394.130	98.937	1066.261	0.000

Sensor pair (start points)	Signed Δx	Signed Δy	Signed Δz	3D distance
ACU_WT06 → ACU_WT07	-0.006	-590.602	0.000	590.602
ACU_WT07 → ACU_FIB04	17.138	-394.130	-17.342	394.884

Source: Annotated_ACU(2).FCStd — CAD extraction / saved annotations. CAD Start_Point positions; ordering and Δu use centerline projection.

19 | Heating tape & cooling-jacket extents

Axial hardware dimensions do not establish the active heat-transfer boundary conditions

Zone	Start u	End u	Axial L	Start s	End s	Outlet clearance
HEATING_TAPE	42.943	1281.193	1238.250	114.235	1352.485	31.750
COOLING_JACKET	215.173	1231.173	1016.000	3129.576	4145.576	82.550

Location	X	Y	Z
JACKET_IN	-1077.480	2434.833	0.000
JACKET_OUT	-70.357	2300.823	0.000
TAPE_IN	-53.182	861.935	0.000
TAPE_OUT	-1280.613	1025.260	0.000

Zone	CAD extraction locator
HEATING_TAPE	Part__Feature146.Face2/Face4; Heating_Tape_Start_Point [DatumPoint006]; support: Part__Feature146.Edge6; Heating_Tape_End_Point [DatumPoint007]; support: Part__Feature146.Edge7
COOLING_JACKET	Part__Feature145.Face7/Face8; Cooling_Jacket_Start_Point [DatumPoint004]; support: Part__Feature182.Edge9; Cooling_Jacket_End_Point [DatumPoint005]; support: Part__Feature181.Edge10

Source: Annotated_ACU(2).FCStd — CAD extraction / saved annotations. Named datums and analytical cylinder / torus faces

20 | Main exit pipe endpoints & cross-sections

TLEP = top-left exit pipe · BREP = bottom-right exit pipe

Exit / endpoint	X	Y	Z
TLEP / start	-1301.784	2461.348	0.000
TLEP / end	-1301.884	2664.548	0.000
BREP / start	0.000	850.900	0.000
BREP / end	0.000	50.800	0.000

Component	OD	ID	Wall thickness	CAD face reference
MAIN_PIPE	33.401	26.645	3.378	Part__Feature140.Face16/Face11; analytical cylinder radii
COOLING_JACKET	158.750	149.200	4.775	Part__Feature145.Face8/Face7; analytical cylinder radii
TLEP_MAIN_TUBE	42.164	32.461	4.851	Part__Feature141.Face6/Face4; analytical cylinder radii
BREP_MAIN_TUBE	42.164	32.461	4.851	Part__Feature142.Face96/Face94; analytical cylinder radii

TLEP axial length = 203.200 mm; BREP axial length = 800.100 mm. Start is loop-side and end is remote-side. These are cylinder-axis limits, not certified pipe/loop junction intersections or expansion-tank connection coordinates. Smaller exit-related PDF diameter callouts must not replace these main-pipe values.

Source: Annotated_ACU(2).FCStd — CAD extraction / saved annotations. Part__Feature141.Face6 and Part__Feature142.Face96.

21 | Solid face boundaries & heater join gap

Individual straight faces use CAD parameter direction, which may oppose the loop-flow convention

Face segment	Axis start X	Axis start Y	Axis end X	Axis end Y	Length
HEATER_LONG	-1255.006	1021.852	-10.614	856.271	1255.360
HEATER_SHORT	-1312.086	1029.448	-1255.435	1021.910	57.150
COOLER_LONG	-45.179	2297.473	-1290.773	2463.214	1256.573
COOLER_SHORT	11.472	2289.935	-45.179	2297.473	57.150
UPCOMER	-1384.915	2380.729	-1384.287	1111.851	1268.878
DOWNCOMER	84.150	939.261	84.150	2206.945	1267.684

Heater: 1255.359980 mm long face + 57.150000 mm short face + 0.433138 mm modeled axial gap = 1312.943118 mm full tangent span. Cooler: 1256.572983 + 57.150000 = 1313.722983 mm; the two faces meet to numerical precision.

Face segment	Source locator
HEATER_LONG	Part__Feature140.Face18
HEATER_SHORT	Part__Feature143.Face13
COOLER_LONG	Part__Feature143.Face17
COOLER_SHORT	Part__Feature140.Face14
UPCOMER	Part__Feature143.Face15
DOWNCOMER	Part__Feature140.Face16

Source: Annotated_ACU(2).FCStd — CAD extraction / saved annotations. Full XYZ axis boundaries, including unrounded Z, are in solid_face_segments.csv.

22 | CAD / text / PDF reconciliation / 1 of 2

Selected geometry always comes from the CAD; legacy values are retained for audit

Parameter	CAD (mm)	Text (mm)	PDF (mm)	Resolution
heater_full_tangent_length	1312.943	1312.900	1255.400	CAD full span selected. PDF is long-face-only; short section and gap omitted.
cooler_full_tangent_length	1313.723	1313.700	1256.600	CAD full span selected. PDF omitted 57.15 mm short section.
upcomer_length	1268.878	1268.900	1268.900	Rounded source values agree with CAD.
downcomer_length	1267.684	1267.700	1267.700	Rounded source values agree with CAD.
top_vertical_axis_offset	1469.065	—	1469.100	CAD selected; projected X offset at upper tangents, not a single perpendicular distance.
bottom_vertical_axis_offset	1468.437	—	1468.400	CAD selected; upcomer is slightly inclined; exact lower offset differs from PDF.
jacket_length	1016.000	1016	1016	CAD selected; rounded source values agree.

Sources: Annotated_ACU(2).FCStd — CAD extraction; ExtractedMeasurements.txt — previous text inputs; ACU_Detailed_General_Arrangement 2.pdf — page 1 dimension / diameter schedule. Row-specific CAD locators and differences are included in comparisons.csv and XLSX Comparisons.

23 | CAD / text / PDF reconciliation / 2 of 2

Selected geometry always comes from the CAD; legacy values are retained for audit

Parameter	CAD (mm)	Text (mm)	PDF (mm)	Resolution
jacket_outlet_clearance	82.550	82.500	25.400	PDF clearance stops at solid seam, not cooler tangent; full clearance is approximately 82.55 mm.
heating_tape_length	1238.250	1238.300	—	CAD exact span selected over rounded TXT.
main_pipe_od	33.401	33.400	33.401	CAD selected.
main_pipe_id	26.645	26.600	26.645	CAD selected; PDF ID rounded to 0.001 mm.
main_tlep_od	42.164	—	1.588	PDF small-tube OD is not the main exit-tube OD; do not transfer that scope.
main_brep_od	42.164	—	15.875	PDF side-probe OD is not the vertical main exit-tube OD; do not transfer that scope.

Sources: Annotated_ACU(2).FCStd — CAD extraction; ExtractedMeasurements.txt — previous text inputs; ACU_Detailed_General_Arrangement 2.pdf — page 1 dimension / diameter schedule.
Row-specific CAD locators and differences are included in comparisons.csv and XLSX Comparisons.

24 | Verification & remaining model questions

CAD global XYZ · all lengths in mm unless noted

ID	Status / topic	Action
B01	OPEN_CAD_MODEL_GAP / Heater solid join	Main centerline span includes this axial separation. Confirm intended assembly/weld gap before treating solids as watertight.
B02	RESOLVED_BY_CAD_SCOPE / Inclined straight lengths	Use full tangent-to-tangent CAD lengths. PDF values are long cylindrical face segments, not complete inclined legs.
B03	RESOLVED_FOR_GEOMETRY / Bend angle convention	Canonical geometry uses actual torus U sweep and major radius. TXT arcs are retained in legacy data, not reused for current stations.
B04	OPEN_INSTALLATION_MAPPING / Sensing point vs CAD annotation	Start/end datum coordinates are now available. The labels do not certify the effective measurement junction or DAQ channel mapping.
B05	OPEN_INTERFACE_TOPOLOGY / Exit interface definitions	Main exit cylinder axis endpoints are extracted. Junction intersections and physical tank ports are not relabeled as these endpoints without checking connections.
B06	PDF_SCOPE_SUPERSEDED / Exit diameter callouts	Use OD 42.164 mm / ID 32.4612 mm for each main vertical exit pipe. Smaller PDF values describe other small-tube/probe scopes.
B07	OPEN_ATTACHMENT_INDEX_REVIEW / PTC 2/3 support edge numbering	Saved PTC_2/3 End_Point coordinates match another circle center in the same component, not the saved support edge number in the extracted BRep order. Preserve saved coordinates and both references; do not relocate to the remote edge.

49 datum-to-circle checks were possible: 47 direct support-index matches and 2 saved points matching another circle edge in the same component. Maximum nearest-circle residual is below 10^{-9} mm. All 221 Part::Feature placements and the parent Part placement were checked as identity before using saved BRep coordinates.

Source: Annotated_ACU(2).FCStd — CAD extraction / saved annotations. Document.xml; annotation_checks.csv; CAD object inventory.

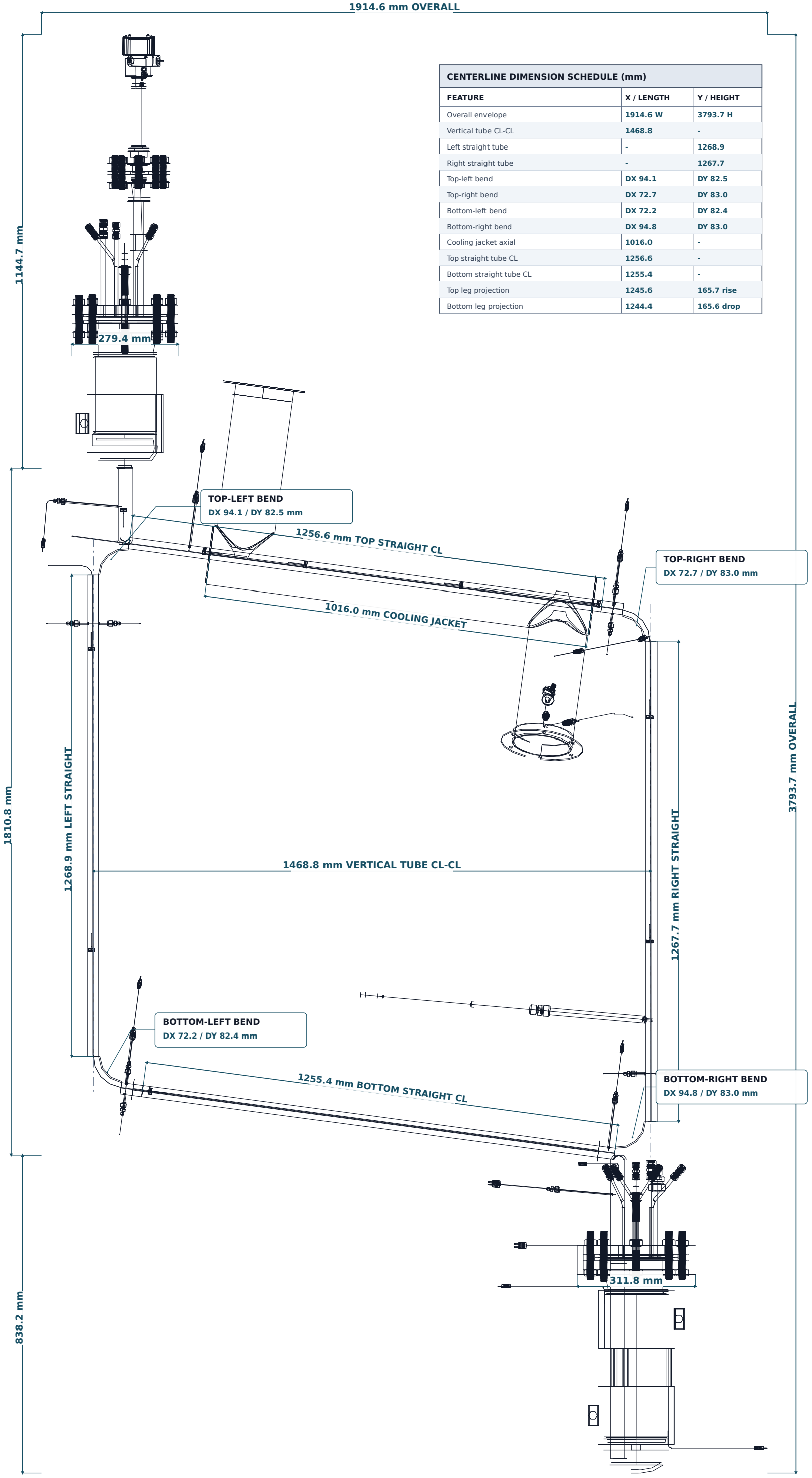
25 | Code access, provenance & retained sources

CAD global XYZ · all lengths in mm unless noted

Start with sensor_points.csv for every saved sensor datum and pipe_segments.csv for the eight flow-ordered loop segments. Use exits.csv for main exit pipes, named_locations.csv for tangent/thermal/join points, and sensors.csv for one row per sensor plus centerline projections.

File / field	Meaning
acu_geometry.json	Single structured dataset. Tables are top-level arrays. metadata defines native frame, units, station origin, source priority, extraction method and limitations.
source_file / source_kind / source_locator	Actual source document filename; CAD extraction or annotation classification; exact object, face or datum reference. Comparisons carry separately named CAD/text/PDF sources.
load_acu.py	Returns pandas DataFrames by table name. Example: from load_acu import load_acu; tables, metadata = load_acu(); xyz = tables["sensor_points"].
validate_acu.py	Checks table counts, endpoint continuity, lengths, projections, source completeness and JSON/CSV consistency.
legacy_*.csv	Revision A text/PDF derivations retained as SUPERSEDED_REFERENCE_ONLY. Do not concatenate them with CAD-primary records.
Nulls and precision	JSON null / CSV blank means undefined, not zero. Numeric exports retain extraction precision; rounded printed values are not manufacturing tolerances.

The next two appendix pages reproduce the original PDF drawings unchanged. Their labels and dimensions remain secondary source evidence; they do not override the preceding CAD-primary sheets.



CENTERLINE DIMENSION SCHEDULE (mm)

FEATURE	X / LENGTH	Y / HEIGHT
Overall envelope	1914.6 W	3793.7 H
Vertical tube CL-CL	1468.8	-
Left straight tube	-	1268.9
Right straight tube	-	1267.7
Top-left bend	DX 94.1	DY 82.5
Top-right bend	DX 72.7	DY 83.0
Bottom-left bend	DX 72.2	DY 82.4
Bottom-right bend	DX 94.8	DY 83.0
Cooling jacket axial	1016.0	-
Top straight tube CL	1256.6	-
Bottom straight tube CL	1255.4	-
Top leg projection	1245.6	165.7 rise
Bottom leg projection	1244.4	165.6 drop

FRONT ELEVATION - CENTERLINE / TANGENT DIMENSIONS

ACU ASSEMBLY - DETAILED GENERAL ARRANGEMENT

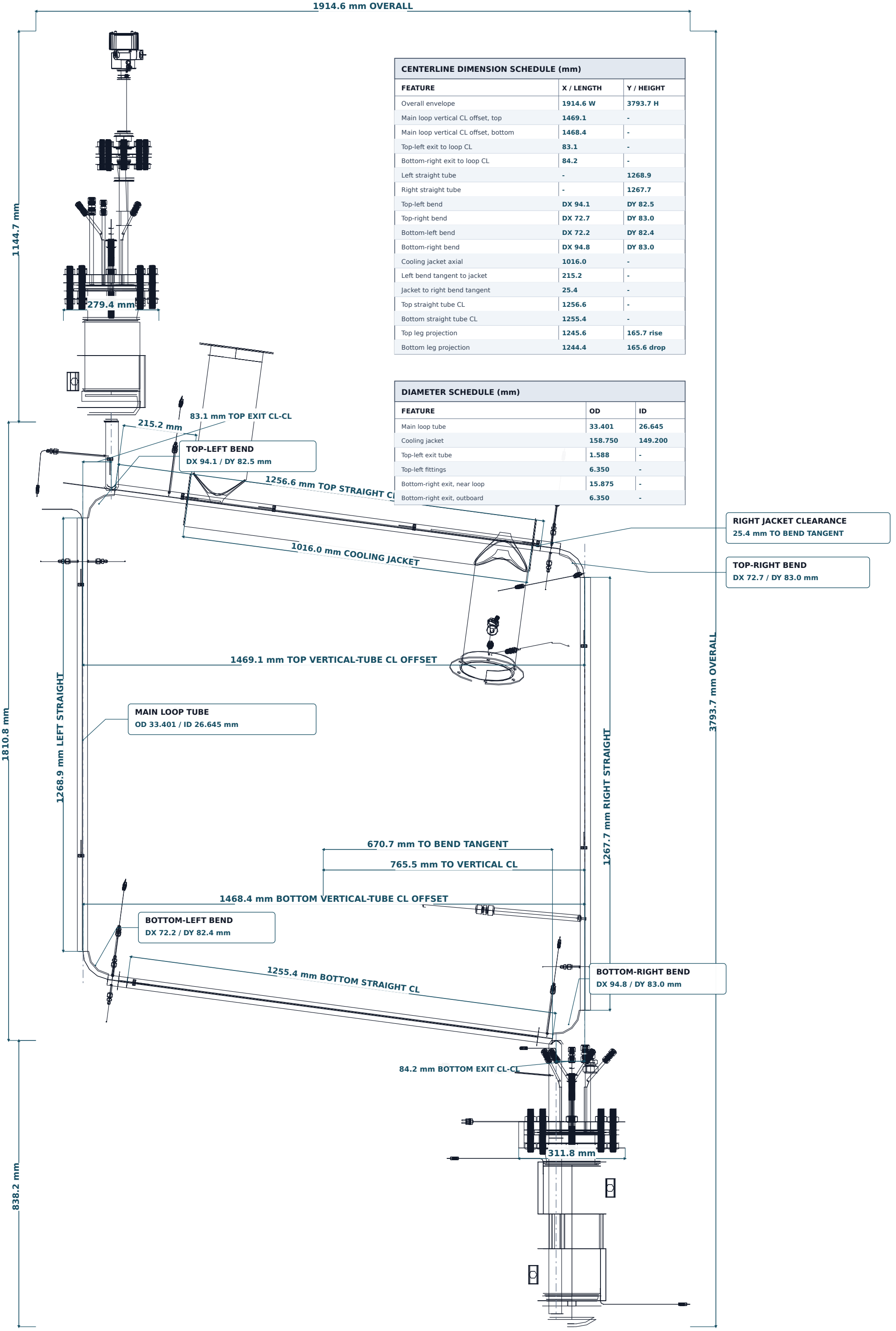
VERTICAL TUBES, BEND OFFSETS, STRAIGHT LEGS, AND COOLING JACKET

BEND DX/DY VALUES ARE PROJECTED TANGENT-TO-TANGENT OFFSETS. VERIFY INTERFACES BEFORE FABRICATION.

SCALE **1:10**

SHEET **A3 / 1 OF 1**

UNITS: mm



CENTERLINE DIMENSION SCHEDULE (mm)

FEATURE	X / LENGTH	Y / HEIGHT
Overall envelope	1914.6 W	3793.7 H
Main loop vertical CL offset, top	1469.1	-
Main loop vertical CL offset, bottom	1468.4	-
Top-left exit to loop CL	83.1	-
Bottom-right exit to loop CL	84.2	-
Left straight tube	-	1268.9
Right straight tube	-	1267.7
Top-left bend	DX 94.1	DY 82.5
Top-right bend	DX 72.7	DY 83.0
Bottom-left bend	DX 72.2	DY 82.4
Bottom-right bend	DX 94.8	DY 83.0
Cooling jacket axial	1016.0	-
Left bend tangent to jacket	215.2	-
Jacket to right bend tangent	25.4	-
Top straight tube CL	1256.6	-
Bottom straight tube CL	1255.4	-
Top leg projection	1245.6	165.7 rise
Bottom leg projection	1244.4	165.6 drop

DIAMETER SCHEDULE (mm)

FEATURE	OD	ID
Main loop tube	33.401	26.645
Cooling jacket	158.750	149.200
Top-left exit tube	1.588	-
Top-left fittings	6.350	-
Bottom-right exit, near loop	15.875	-
Bottom-right exit, outboard	6.350	-

FRONT ELEVATION - CENTERLINE / TANGENT DIMENSIONS

ACU ASSEMBLY - DETAILED GENERAL ARRANGEMENT

VERTICAL TUBES, EXIT CENTERLINES, BEND OFFSETS, DIAMETERS, AND COOLING JACKET CLEARANCES

TUBE OFFSETS ARE PROJECTED HORIZONTAL CENTERLINE DISTANCES AT THE STRAIGHT-TUBE TANGENTS. VERIFY BEFORE FABRICATION.

SCALE **1:10**

SHEET **A3 / 1 OF 1**

UNITS: mm